

Transform Gut Motility

CLINICAL APPLICATIONS

- Supports Gastrointestinal Function
- Supports Bowel Regularity

OVERVIEW



Cape aloe has been used for thousands of years to promote the health of epithelial tissue, including the bowel lining. Cape aloe contains a class of phytonutrient compounds called anthraquinone glycosides. When ingested, these compounds are activated through bacterial metabolism in the gut, promote water retention in the stool, and stimulate peristalsis in the large intestine. By causing active chloride secretion, cape aloe increases the water and electrolyte content in the stool and eases elimination without artificial additives and cramping associated with other therapies.

Super Aloe is intended for the relief of occasional constipation. Cape aloe, the solid residue obtained by evaporating liquid aloe ferox, has been shown in the research to affect regularity due to its influence on colon peristalsis, which it triggers to contract, resulting in accelerated intestinal passage. This formulation Super Aloe is available as 250 and 450 mg capsules, two potencies to provide tailored natural support for elimination.

Suggested Use:

1 capsule at bedtime (allow 24 hours before redosing) or as recommended by your health care professional

INGREDIENT BENEFITS

Cape Aloe

Despite the lack of published studies, aloe preparations have been in the United States Pharmacopoeia for more than 100 years. Anecdotal evidence suggests cape aloe supports normal inflammatory response in the gastrointestinal tract. Studies demonstrate that cape aloe increases stool bulk and bowel transit time and improves microflora balance. Cape aloe also supports the colon to help maintain a healthy gastrointestinal tract.

Collagen Factors



CLINICAL APPLICATIONS

- *Maintains Healthy Joint Function for Normal Mobility and Lubrication*
- *Provides Precursors for Generation of Cartilage, Tendons, Ligaments, Fascia and Skin*
- *Promotes a Healthy Joint Environment by Supporting Normal Activity of Cytokines and Catabolic Enzymes*

MUSCULOSKELETAL HEALTH

Collagen Factors provides targeted precursors and cofactors known to be involved in the biochemical processes that support healthy connective tissue in the skin, tendons, ligaments, cartilage, joints, and bone extracellular matrix. This formulation contains patented hyaluronic acid, MSM, free-form amino acids, key minerals and vitamin C.

Hyaluronic Acid Extract[†]

Mobilee® is a patented, high-molecular-weight hyaluronic acid (HA), which enhances the viscoelastic, lubricating properties of synovial fluid. In addition, it contains collagen and other glycosaminoglycans (GAGs) to enhance skin and joint health. Hyaluronic acid also plays a role in the biophysical, biochemical and cell regulation processes in joint synovial tissues. Scientific evidence shows that Mobilee® supports chondrocytes and synovial cells function, and is two to four times more active than fermented sources of HA in nourishing and supporting the health of synovial fluid.^{1,2} The latest clinical research includes HA in proactive and maintenance approaches to joint care that supports normal range-of-motion. Research suggests that Mobilee® supports the quality of synovial fluid by positively influencing synovial HA concentration, and by reducing the expression of degradative factors in synovial fluid.³⁻⁶

Specific Precursors[†]

Collagen is comprised of and formed from the amino acids proline and lysine. Lysine is an essential amino acid the body relies on adequate dietary intake for its involvement in the synthesis of connective tissues. Proline is well-known for its involvement in the triple helix of collagen—it is regularly

spaced with glycine in the amino acid sequence of collagen fiber, which together comprise about 50% of total collagen sequencing. Proline contributes to not only collagen formation, but muscle, tendons and skin. Lysine helps support the linking and stabilization of collagen, and contributes to the collagen matrix formation of veins, arteries and capillaries. During procollagen synthesis, both proline and lysine are hydroxylated to hydroxyproline and hydroxylysine, a reaction requiring vitamin C as a cofactor.⁹⁻¹¹ Inadequate levels of vitamin C can impact this reaction, resulting in incomplete formation of the helical structure, in which leads to a delicate and easily destroyed structure. Vitamin C is combined with two amino acids—lysine and proline—to form procollagen. Procollagen is then used to manufacture one of several types of collagen found in different tissues throughout the body.¹²⁻¹³ Furthermore, an animal study suggests that serum ascorbate levels influence fluid accumulation in the joint, reinforcing the importance of this cofactor in joint health.¹²

Targeted Cofactors

Zinc is an essential trace mineral that supports DNA synthesis, cell division, cell membrane structure, immune function, and protein synthesis. Requirements are highest when the body's repair system is most active. **MSM** is responsible for the flexible disulfide bonds between cells, including those that make up connective tissue and the skin. It blocks undesirable chemical and physical cross-linking of collagen and preserves connective tissue pliancy, which is crucial for supporting healthy collagen. MSM has been shown to significantly increase plasma glutathione levels, an additional benefit in maintaining collagen's defensive mechanisms against oxidative stress.

[†] These statements have not been evaluated by the Food and Drug Administration. This product is not intended to diagnose, treat, cure, or prevent any disease.

Multiple clinical studies also show that MSM, in combination with other joint health ingredients, supports tendon and joint health.⁶⁻⁷ **Manganese** assists the growth and development of normal bone and the synthesis of cartilage. Manganese is also a structural component of antioxidant enzymes, and aids in amino acid production. During the tissue synthesis process, which requires an increased production of collagen, manganese is required for the activation of prolydase, an enzyme that functions to provide the amino acid proline, for collagen formation in human skin cells.¹⁵ Nutritional co-factors required for tissue regeneration include zinc, vitamin C and amino acids. Inadequacy of these nutrients impacts endogenous collagen production and connective tissue synthesis.

Directions

2 capsules per day or as recommended by your health care professional.

Does Not Contain

Gluten, yeast, artificial colors or flavors.

Cautions

If you are pregnant or nursing, consult your physician before taking this product.

Supplement Facts ^{V1}		
Serving Size 2 Capsules		
Servings Per Container 30		
	Amount Per Serving	% Daily Value
Vitamin C (as Ascorbic Acid USP)	100 mg	111%
Zinc (as TRAACS™ Zinc Bisglycinate Chelate)	15 mg	136%
Manganese (as TRAACS™ Manganese Bisglycinate Chelate)	10 mg	435%
Methylsulfonylmethane (MSM) (OptiMSM®)	500 mg	*
L-Lysine Hydrochloride USP	200 mg	*
L-Proline USP	200 mg	*
Mobilee® (Standardized to contain 40 mg Hyaluronic Acid)	80 mg	*
* Daily Value not established.		

ID# 332060 60 Capsules

References

1. Torrent A, Ruhí R, Theodosakis J, et al. Comparative efficacy of IB0004, extracted hyaluronic acid (HA) and fermented HA on the synthesis of endogenous HA by human synoviocytes. *Osteoarthritis Cartilage*. 2009;17(Suppl 1):S278-79. – 10x HA secretion
2. Torrent A, Ruhí R, Martínez C, et al. Anti-inflammatory activity and absorption of a natural rooster comb extract (Hyal-Joint®). *Osteoarthritis and Cartilage*. 2010 Oct;18(Suppl 2):S246-47. doi:10.1016/S1063-4584(10)60577-8. – reduction of inflammation
3. Möller I, Martinez-Puig D, Chetrit C. Oral administration of a natural extract rich in hyaluronic acid for the treatment of knee OA with synovitis: a retrospective cohort study. *Clinical Nutrition Supplements* 2009;4(2):171-172
4. Martinez-Puig D, Möller I, Fernández C, Chetrit C. Efficacy of oral administration of yoghurt supplemented with a preparation containing hyaluronic acid (Mobilee™) in adults with mild joint discomfort: a randomized, double-blind, placebo controlled intervention study. *Mediterranean Journal of Nutrition and Metabolism* 2013;6:63–68.
5. Sánchez J, Bonet ML, Keijer J, van Schothorst EM, Möller I, Chetrit C, Martinez-Puig D, Palou A. Blood cells transcriptomics as source of potential biomarkers of articular health improvement: effects of oral intake of a rooster combs extract rich in hyaluronic acid. *Genes & Nutrition* 2014; 9: 417
6. Nieman DC, Shanely RA, Luo B, Dew D, Meaney MP, Sha W. A commercialized dietary supplement alleviates joint pain in community adults: a double-blind, placebo-controlled community trial. *Nutr J* 2013;12(1):154.
7. Brien S, Prescott P, Lewith G. Meta-analysis of the related nutritional supplements dimethyl sulfoxide and methylsulfonylmethane in the treatment of osteoarthritis of the knee.
8. Brien S, Prescott P, Lewith G. Meta-analysis of the related nutritional supplements dimethyl sulfoxide and methylsulfonylmethane in the treatment of osteoarthritis of the knee. *Evid Based Complement Alternat Med*. 2011;2011:528403. doi: 10.1093/ecam/nep045. [PMID: 19474240]
9. S Murad, D Grove, K A Lindberg, G Reynolds, A Sivaraja, S R Pinnell. Regulation of collagen synthesis by ascorbic acid. *Proc Natl Aca Sci USA*. 1981 May; 78(5):2879-2882
10. Boyera N, Galey I, Bernard BA. Effect of vitamin C and its derivatives on collagen synthesis and cross-linking by normal human fibroblasts. *Int J Cosmet Scie*. 1998 Jun;20(3):151-8. Doi: 10.1046/j. 1467-2494.1998.171747.x.
11. Andreas Herchenhan, Franziska Uhlenbrock, Pernilla Eliasson, et al. Lysyl Oxidase Activity Is Required for Ordered Collagen Fibrillogenesis by Tendon Cells. *J Biol Chem*. 2015 Jun 26;290(26):16440-16450.
12. Simões SI, Eleutério CV, Cruz ME, et al. Biochemical changes in arthritic rats: dehydroascorbic and ascorbic acid levels. *Eur J Pharm Sci*. 2003;18(2):185-89. [PMID: 12594012]
13. Li P, Wu G. Roles of dietary glycine, proline, and hydroxyproline in collagen synthesis and animal growth. *Amino Acids*. 2017 Sep 20.
14. Frey J, Raby N. Lysine and collagen. *Annales de Biologie Clinique*. 01991, 49 (1):36-39.
15. Muszynska A, Palka J, Gorodkiewicz E. The mechanism of daunorubicin-induced inhibition of prolylase activity in human skin fibroblasts and its implication to impaired collagen biosynthesis. *Exp Toxicol Pathol*. 2000 May;52(2):149-55.

LT-PDN-203-A